SOFTWARE ENGINEERING INTERVIEW — QUICK GUIDE

CodeTrackr

A plain-English 20-page summary of the full technical prep document — read once or twice and you can hold your own in an interview

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Condensed from `CodeTrackr_Interview_Preparation.md` (64 pp)

Full versions in `docs/interview-preparation/`

REAL = in the code now · **BETTER** = a suggested improvement

Contents

1. What CodeTrackr is (in one breath)

CodeTrackr watches how you code and shows you the numbers — built so a friend group can keep **friendly competition** going.

The motive: my friends and I run coding contests and daily-practice streaks, and everyone always *claims* they put in the hours. CodeTrackr makes it automatic: make a **group**, everyone installs the extension, and the group page shows who actually coded, in what languages, and (because it tracks failed commands/builds too) roughly who fought the most errors.

There are three parts:

1. **A VS Code extension.** It runs quietly in your editor. It notices what file you're in, what language, how much you type, how many terminal commands you run **and whether they passed or failed**, when you commit — and every ~2 minutes it sends a small summary to a server.
2. **A backend server** (Node + Express + MongoDB). It checks who the data belongs to using a personal key, then **merges the summary into a 10-minute "bucket" record** (many summaries → one row), and does the maths when the website asks for it.
3. **A React website** (the dashboard). You log in with Google. It shows charts of your coding, **groups with a per-member leaderboard** (the main feature), a global leaderboard, goal tracking, and an "Insights" page with things like your most productive two-hour window.

Two different logins:

- The **website** uses Google sign-in → the server gives you a JWT stored in a cookie.
- The **extension** uses a **64-character API key** you copy from the website.

Where it runs: website on Vercel, server on Render, database on MongoDB Atlas.

The one honest thing to say first

The "Insights" page is **not machine learning**. It's ordinary maths — averages, medians, a "how steady are you" score. No model, no training, no AI. That was a deliberate choice so every number can be explained. (More in section 9.)

2. Three ways to explain it

30 seconds

> *"CodeTrackr is a coding-activity tracker built around friendly competition. My friends and I wanted to see who was actually grinding during contest weeks, so I built a VS Code extension that records what you work on — time per language, editor activity, terminal commands and whether they passed or failed, git commits — and sends it to my backend. You make a group with your friends and the app shows a per-member leaderboard plus a global one, goal tracking, and an insights page. MERN stack plus a TypeScript VS Code extension."*

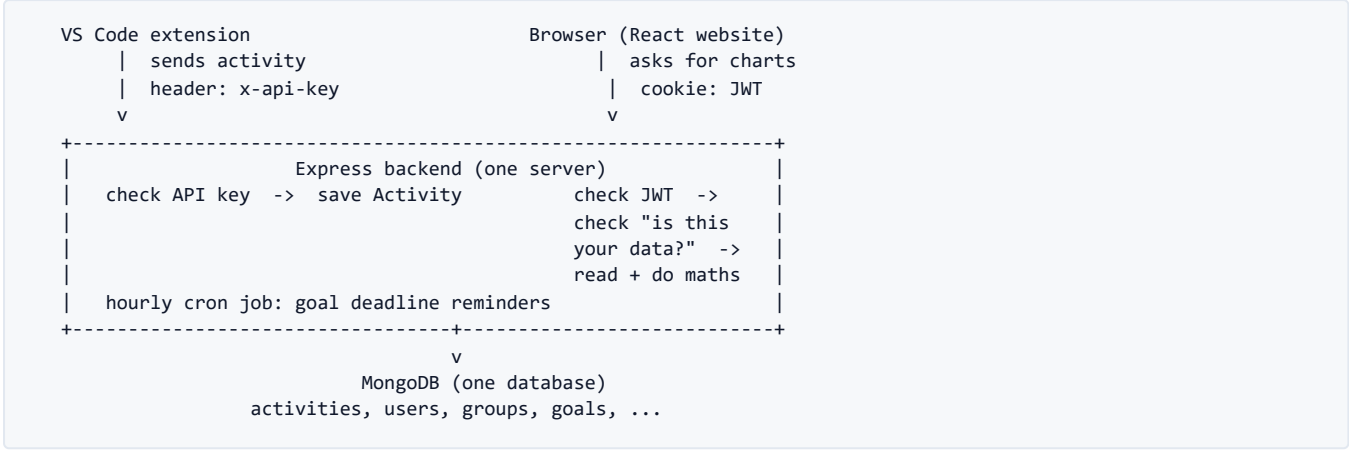
1 minute

> *"The idea came from my friend group — during coding contests we kept arguing about who'd done the work, so I built something to measure it. My VS Code extension hooks into editor, terminal, git and window events and keeps running counters — lines added, 'churn' (code you wrote then deleted), focused minutes, command and build success rates. It sends those to an Express API, checked by a per-user API key. The backend merges each summary into a 10-minute record in MongoDB. The React frontend shows daily/weekly Chart.js graphs;* ***groups you create with a per-member leaderboard****; a global leaderboard; goals on a calendar with cron-driven reminders; and an insights page that works out confidence-gated productivity metrics using plain statistics."*

2 minutes

> *Add this to the 1-minute version:*
>
> *"The interesting design problem was linking the extension to the account without a full login screen inside VS Code. I used an API-key model: on first sign-in the server generates a random 64-character key and stores it on the user. The extension sends it as a header; the server looks up the user by that key. It works, but in a review I'd call it what it actually is — a password-like credential that never expires and is stored as plain text — and I can explain how I'd harden it.*
>
> *The other thing I'm proud of is the accuracy work. The first version measured time as 'editor was open' and lines as the net change in line count, which is zero for any refactor. I rewrote the trackers to count gross edits and 'churn', and to only count focused time. I also audited my own code into an IMPROVEMENT_PLAN file with about 30 findings and fixed the serious security ones, with a test that fails if any protected route loses its auth check."*

3. The big picture (architecture)



What kind of architecture is this? Say: *"Client–server, REST-ish, a modular monolith."*

- **Client–server:** thin clients (extension, website), one server that owns the truth.
- **Monolith:** one Node process. Features are split into separate route files but deploy together. This is fine for one developer and one database.
- **Modular:** each feature (analytics, groups, goals, leaderboard) is its own file, so you could split one out later. The obvious first candidate is the activity-ingest endpoint.
- **No** message queue, **no** Redis cache, **no** WebSockets. Notifications are fetched by the browser every 30 seconds (polling).

Why not microservices? One developer, one database, and all the features share the same data. Splitting them would add network calls and complexity for no benefit yet.

4. The VS Code extension

How it starts and runs

When VS Code finishes starting up, the extension wakes (`onStartupFinished`), creates five small "trackers", and starts a timer. The timer ticks every **30 seconds**: if you've been idle 2+ minutes it pauses (resuming when you type again); if you've been active for at least **2 minutes** (the default, `minFlushMinutes` — changed from 30 seconds in v2.3.0) **and** something actually happened (edits / commands / commits), it sends one summary. A flush with no real activity is held so the time carries to the next one.

What it records

Each tracker keeps counters, then hands them over ("snapshot and reset"):

Tracker	What it counts
Editor	characters and lines typed/deleted, **churn** (wrote then deleted within 10 min), undo/redo, saves, file switches, time reading vs writing
Focus	minutes the window was actually in front, and "flow blocks" (a work stretch that ends after 2 min idle)
Git	commits (via VS Code's built-in Git, so it catches GUI commits too), uncommitted file count
Terminal	commands run, which succeeded/failed, build/test runs, repeated failures
Debug	number of debug sessions

What it sends (the payload)

One JSON object per flush: timestamp (set to the **start** of the interval, not "now", so hour-of-day charts aren't skewed), file **name only** (never the full path), file type, project, language, duration in **seconds**, line counts, and the four tracker summaries.

Privacy: it never sends file contents, diffs, commit messages, or folder paths. Terminal commands are cleaned (`token=...` becomes `token=***`) and shortened.

What happens when things go wrong

Situation	What happens now (**REAL**)
You go idle	It pauses, then resumes on your next keystroke
VS Code closes	It tries one last send, then stops
No internet	The unsent minutes stay **in memory** and retry next tick. If VS Code restarts first, that time is lost — **there's no on-disk queue**
Key is wrong / was reset	You get a one-time popup: "Set API Key" / "Open Dashboard"
Same data sent twice	The backend adds it to the **same 10-minute bucket** (not a new row), so it double-counts *within* that bucket — still not fully idempotent

Where the key is stored on your machine: in VS Code's settings file, as **plain text**. **BETTER:** use VS Code's SecretStorage (the OS keychain).

5. The API key (this is the #1 interview topic)

How it works (**REAL**)

1. You sign in with Google for the first time.
2. The server creates your user and runs `crypto.randomBytes(32)` → a 64-character key.
3. It's saved on your user record as **plain text**.
4. The website shows it to you. You paste it into the extension.
5. The extension sends it on every request as a header: `x-api-key: <key>`.
6. The server does `User.findOne({ apiKey })` to find you, then saves the activity under your ID.
7. If you think it leaked, you click "Regenerate" — a new key is made, the old one dies.

"Is it just an identifier?" — say NO

An **identifier** (like a username) is safe to show publicly. This key is different — holding it lets you **write data**. So it's a **credential**, not an identifier. More precisely:

- It **identifies** you (one key = one user).
- It **authenticates** you (no second check — having it is enough).
- It's a **bearer token** (whoever holds it can use it).
- It **never expires** and has **no limits** on what it can do.
- It's stored and compared as **plain text**.

Say this line: *"It's basically a password that never changes, stored in plain text. That's an OK choice for a first version — it's one database lookup and works offline — but I'd call it a credential in a review, and I know how to make it safer."*

What an attacker could do

With a stolen key they can **fake activity** for you — send fake hours in a loop and mess up the leaderboard and your charts. They **cannot** read your dashboard (that needs your login cookie). There's no rate limit and no sanity check on the numbers, so it's easy.

How to fix it (**BETTER**)

- **Store a hash, not the key.** Give the key as `ct_<id>_<secret>`, look up by the `id`, compare the `secret` against a hash. A database leak then reveals nothing usable.
- **Allow several keys per user**, each named ("laptop", "desktop") and revocable on its own.
- **Rotation with a grace period** — the old key still works for 24 hours after you make a new one, so tracking doesn't suddenly break.
- **Even better:** the website hands out a short-lived signed token the extension refreshes, or use an OAuth "device flow" so the extension never holds a long-term secret.

6. The database (MongoDB)

The collections

Collection	What it holds	Notes
activities	one record per **10-minute window** (per user + project + language) — changed 2026-09-08; was one per flush	the big one; a 400-day auto-delete rule now caps its growth
dailysummaries	one row per user per day, built by a nightly job from `activities`	infrastructure for future "all-time" reads — nothing uses it yet
users	Google ID, email, name, the **plain-text** API key	
groups + **groupmembers**	competition groups + who's in them	`groupmembers` is a proper join table with a "one row per (group, user)" rule
goals	title, target hours, tech stack, deadline, status	**nothing in the code ever marks a goal "completed"** — only the demo seed script does
teams	older idea; members stored inside the team record	backend exists but **the website never shows it** — dead code
notifications	goal deadline reminders	created by the cron job

The `activities` record

Key fields: `userId` (stored as **text**, not a real reference — a quirk), file/language/ project, `duration` in **seconds** (added up as real measured seconds — the totals don't change from bucketing), line counts, `timestamp` (= the 10-minute window start), `files[]`, `flushCount`, and four sub-objects (`terminalAnalytics`, `editorAnalytics`, `focusAnalytics`, `gitAnalytics`) that now only store the numbers that actually happened (a dead `date` field was removed).

Why MongoDB (a good answer)

- Activity records are **written once, never edited** (each 10-minute bucket is just added to with `$inc`) — good fit for a document store.
- The shape **grew over time** — I added three tracker sub-objects with **no migration**, because Mongoose just treats missing fields as zero.
- The main read is "give me one user's records for the last week, then add them up" — no joins needed on the hot path.
- Free managed hosting (Atlas).

When SQL (Postgres) would be better

- The **relational bits** — group membership, goal ownership — where foreign keys and "delete cascades" would remove defensive code.
- The **leaderboard** — that's a `GROUP BY user` which SQL does with an index, instead of loading the whole table into Node.
- **Transactions** for multi-step actions.

Indexes (updated 2026-09-08)

`{userId, timestamp}` (the one the read queries actually need — added), `{userId, project}`, `{userId, language}`, a **unique** `{userId, project, language, bucketStart}` (makes the 10-minute merge race-safe), and a **400-day auto-delete** rule on `createdAt`. The old `{userId, date}` index was dropped along with the `date` field. **Still missing:** anything that would help the global leaderboard — but that's a full table scan by design; the fix is a running-totals table, not an index.

7. The backend and API

How one request flows

Extension sending data: `POST /api/extension/track` → check the API key → basic checks (file, language, duration present) → clean the numbers → work out which 10-minute window this belongs to → **add the numbers into that window's record** (`findOneAndUpdate` with `$inc` , create it if missing) → return 201. (Set `ACTIVITY_BUCKET_MS=0` and it goes back to one new row per flush.)

Website reading data: `GET /api/analytics/:userId` → check the JWT cookie → **check "is this your own data?"** (403 if not) → `Activity.find(last 7 days)` → add it up in JavaScript → return JSON → Chart.js draws it.

The main endpoints

What	Endpoint	Login	Notes
Send activity	`POST /api/extension/track`	API key	one record per call
Verify key	`GET /api/extension/verify`	API key	used during setup
Daily / weekly charts	`GET /api/analytics/:userId` `/weekly/:userId`	JWT + ownership check	adds up in JavaScript
Insights	`GET /api/metrics`	JWT	**no `:userId` in the URL** — uses your session only, so it can't leak someone else's data
Leaderboard	`GET /api/leaderboard`	JWT	reads the whole `activities` table every time
Groups	`/api/groups/*`	JWT (+ membership check for details)	
Goals	`/api/goals/*`	JWT (+ owner check)	
Notifications	`/api/notifications/*`	JWT	correctly scoped to you
Google login	`/auth/google` , `/auth/google/callback`	—	sets the JWT cookie

Weak spots in the API

- Every route does its own `try/catch` and returns `error.message` to the client — that used to **leak internal details** — ✅ fixed 2026-09-09 with one central `(err,req,res,next)` handler (correlation id, generic body).
- Rate limiting on `/auth` (50/15min) and `/api/extension` (120/min) via `express-rate-limit` (added 2026-09-09); other routes (e.g. group `/join`) are still unlimited.
- Security headers via `helmet()` (added 2026-09-09).
- **No real input validation** — `duration: 999999999` is accepted; you can even send your own `timestamp` .
- No pagination on the leaderboard.

8. The frontend (React)

- **React 19 + Vite + TypeScript + Tailwind.** `react-router-dom` for pages.
- `App.tsx` checks `GET /api/user/profile` on load. If that works, show the app. If not, show the login page.
- **State:** just `useState` in each page, plus passing the `user` object down as a prop. The only React Context is the **theme** (28 colour themes, saved in `localStorage`).
- **No data-fetching library.** `@tanstack/react-query` is installed but **never used**, so every page reload re-fetches everything. **BETTER:** actually use React Query.
- **Charts:** Chart.js (Line, Pie, Bar).

Things to know (and own honestly)

- `npm run build` **is green** as of 2026-09-09 — the TypeScript check had 29 old errors (mostly unused imports). `vite build` alone works. It's a cleanup job, not a design flaw.
- The dashboard's **"Repeated Failures" panel** now shows the real `repeatedFailedCommands` (fixed 2026-09-09; it used to render fake hard-coded data).
- The **Goals page to-do list isn't saved** — it only lives in React state until you refresh.
- The **Teams page isn't linked** anywhere.

9. Insights — "is this machine learning?"

No. Say it plainly. It's plain statistics, calculated fresh every time you open the page.

The metrics (simple definitions)

Metric	What it means	How it's worked out
Deep work ratio	how much focused time was in long stretches	minutes in blocks ≥ 25 min ÷ total focused minutes
Flow blocks	the shape of your sessions	median, longest, and count of your work stretches
Consistency	how steady you are day to day	`1 − (how spread out your daily minutes are)` — steady beats bursty
True peak hour	your *most productive* hour, not your busiest	the hour with the best score of `commits + lines − churn`
Estimation accuracy	do you underestimate goals?	average of (hours you actually did ÷ hours you guessed)

Key points for the interview

- **Runs on every page load. Not cached. BETTER:** cache it per (user, time window).
- **Needs recent data.** If you're on an old extension version, the focus metrics show "—" instead of "0%".
- **Why statistics instead of ML?** No labelled outcomes to learn from, not enough users, and every number needs to be explainable. Statistics work from day one.
- **The AI part is designed but not built.** There's a plan for an LLM that would *describe* these numbers in sentences — with a strict rule that it can't invent any number — but no code for it yet.

If asked "how would you add real ML?"

Build feature vectors per session (focused minutes, churn ratio, commits, hour of day), normalise per user, then start with **unsupervised clustering** (k-means) to find session types like "deep focus" vs "firefighting". Only do supervised learning once you actually store outcomes like "goal missed". Serve it from a background job, never inside the request.

10. The social features — groups (the core) and the global leaderboard

Groups are why the project exists — friendly competition in a friend group. You make a group for a contest week or daily practice, everyone joins, and because everyone's editor is tracked automatically the group page ranks who actually did the work.

The global leaderboard — how ranking works (**REAL**)

Rank = **total coding hours, all time, highest first**. Ties keep their existing order.

How it's calculated

Every time someone opens the page:

1. Load **every user**.
2. Run an aggregation over the **entire** `activities` **collection** — sum hours and lines per user.
3. Merge and sort **in Node**.
4. Give each user scores (speed, quality, etc.) **relative to the current top user** — so everyone's score shifts when the leader codes more.

The problems

- **It reads the whole activity table on every request.** No time window, no cache, no pagination. This is the **first thing that breaks** as data grows.
- It returns **every user's email**.
- "Commits" on the board is actually **record count**, not real git commits (mislabelled).

The fix (**BETTER**) — know this well

1. Keep a small `UserStats` **table**: one row per user with running totals, updated with `$inc` whenever activity comes in.
2. The leaderboard becomes `UserStats.find().sort().limit(50)` — reads 50 rows, not millions.
3. For "what's my rank?" instantly, use a **Redis sorted set**: `ZADD` on update, `ZREVRANGE` for a page, `ZREVRANK` for your position — all O(log n).
4. Add time-windowed tables for "this week" boards. Paginate. Cache the top 50.

Groups

- **Create:** name, description, public/private. Private groups get a **hashed** (scrypt) password. The creator becomes the first member.
- **Join:** public = one click; private = enter the password (old plain-text ones still work and get upgraded to a hash on the next correct login).

- **Membership** is a `groupmembers` join table with a "one row per (group, user)" unique rule — a solid race protection.
- **View details:** members-only (403 otherwise), then a **per-member leaderboard** of coding hours + lines added (same slow full-scan pattern as the global one).
- **Leave:** the last member leaving deletes the group.

The gap vs the original idea: I wanted the group view to also show **how many errors each person hit**. The extension *does* record every member's failed commands / failed builds / repeated failures — the data's there — but the group leaderboard currently ranks by hours + lines only. Adding the error/build-success columns is a read-path change (extend the same `$group` with `$sum` of the failure counters) and is the top thing on my list. Same for scoping the board to a contest week (`?from=&to=` + a `$match` on `timestamp`).

Other group weak spots: no admin/owner powers (`createdBy` stored but unused — no kick or rename); a double-join returns **409** (fixed 2026-09-09; the handler detects `11000`); "discover" lists private groups too (password-gated, not hidden); no rate limit on join, so passwords can be guessed.

11. Login and security

The website login flow

1. Click login → go to `/auth/google` → Google → back to `/auth/google/callback`.
2. The server finds or creates your user, then signs a **JWT** with your ID, name, email. Valid for **1 day**.
3. The JWT goes into an **httpOnly cookie** (JavaScript can't read it).
4. Every protected route reads the cookie, verifies the JWT, loads the user.
5. Logout clears the cookie.

Login weak spots

- The signing secret is required at boot — the app refuses to start without `JWT_SECRET` (fixed 2026-09-09; was an unsafe fallback).
- **No refresh token** — after 1 day you're just logged out.
- **No way to revoke** a token early (no server-side session list).
- The cookie is `sameSite: 'none'` in production with no CSRF token — low impact here because most actions are POSTs with a JSON body, but worth mentioning.

Security — already fixed (on the current branch)

- Analytics endpoints now check **"is this your data?"** (was: anyone with your ID could read your history).
- Old **unauthenticated** write/read endpoints **deleted**.
- Group passwords are now **hashed** (were plain text).
- The `AUTH_BYPASS` dev flag is **refused in production**.
- Goal and team endpoints now check ownership/membership.
- There's a test that **statically scans the routes** and fails if any protected route loses its auth middleware.

Security — still open (be honest about these)

Issue	Why it matters
API key is plain text, never expires, no limits	leak = someone fakes your activity
Rate limiting only on `/auth` + `/api/extension` (2026-09-09)	group `/join` password guessing still unlimited
~~No security headers~~ ✅ `helmet()` added 2026-09-09	HSTS, `nosniff`, no `X-Powered-By`
~~No input validation on activity~~ ✅ 2026-09-09	`duration` capped at 3600, length/timestamp bounds; still no idempotency key
~~Errors return `error.message`~~ ✅ fixed 2026-09-09	central handler, generic body + id
Leaderboard shows every email	privacy
No duplicate/replay protection	resend a request → counted again

Issue	Why it matters
Group search puts user text straight into a regex	slow-regex (ReDoS) risk

"Can users cheat the leaderboard?" — yes, easily. `curl` the track endpoint with `duration: 3600` in a loop using a valid key; one call = one fake hour. Fix: validate the duration (1–3600s), rate-limit per key, require a unique ID per send, and only trust hours that also have editor/focus/git activity.

12. Scaling up — 10k, 100k, 1M users

Users	What happens	What to do
10,000	Mostly fine. Writes are now ~10× lower (10-minute buckets), and the `timestamp` index + 400-day auto-delete are in. The leaderboard still gets slow once `activities` is millions of rows.	Repoint the leaderboard / all-time reads at the `dailysummaries` rollup that already exists; put a 30-day window + limit on the leaderboard query.
100,000	Leaderboard is unusable without a running-totals table. No caching means the same maths runs over and over.	Build the `UserStats` rollup (`$inc` on write). Move the JS-side analytics into MongoDB `$group` .
1,000,000	One server + one database is the wrong shape.	Put a **queue** in front of activity ingest → workers write to a **sharded** `activities` (split by userId) plus rollups. **Redis** for the leaderboard and the insights cache. Serve charts from precomputed rollups. Archive old raw records. Add real monitoring.

The one-line answer: *"Yesterday's change — merging flushes into 10-minute buckets on write — cut the write and row rate about 10× with no change to the numbers. The next bottleneck is still the leaderboard, because it reads the whole activity table per request; the fix is a per-user running-totals table updated on write, turning a full-table scan into a 50-row read, and I already built the daily-summary collection it would sit on."*

13. Testing and deployment

Testing — the honest picture

What exists: small test scripts using Node's built-in `assert` (no framework).

- **Backend:** 10 files, ~104 checks — the streak logic, the number-cleaning functions, the five insight formulas, the ownership helper, the password hashing, the route scanner that enforces auth, and (2026-09-08) the bucketing maths, the schema/index shape, the ingest wiring, and the daily-rollup builder.
- **Extension:** 2 files — the trackers' behaviour, and an "activation" test that loads the real built bundle and checks every command is registered and no network call happens without a key.

What's missing:

- **No integration tests** — nothing starts the server and hits a real database.
- **No frontend tests at all.**
- Nothing has ever run against a real database — the queries are only reasoned about.

BETTER: add `supertest` + an in-memory MongoDB for API tests; React Testing Library for the frontend; one end-to-end test with Playwright.

Deployment

- **Website:** Vercel. Build is `tsc -b && vite build` (✅ green as of 2026-09-09).
- **Backend:** Render (a normal always-on Node process). There's also a Vercel serverless config in the repo.
- **Database:** MongoDB Atlas.
- **Extension:** packaged with `vsce`, published to the VS Code Marketplace.
- GitHub Actions CI on push (2026-09-09): backend + extension tests, frontend build. Still no CD, no Dockerfile.

One gotcha: the deadline-reminder cron job uses `node-cron`. That works on Render (always on) and — since 2026-09-09 — is serverless-safe: `initScheduler()` / `app.listen()` are behind a `require.main` guard, and an external trigger hits `POST /api/internal/run-*`. Before the fix the hourly job never fires. **BETTER:** use Vercel Cron or an external scheduler hitting a protected endpoint.

14. Quick Q&A (memorise these)

Why MongoDB? Activity records are write-once and the shape changed over time — I added three sub-objects with no migration. Main read is "one user's week, then sum". No hot-path joins.

Why not Postgres? Honestly, the relational parts (groups, goals) and the leaderboard would be better in SQL. The activity stream suits documents. If I redid it with the social features front of mind, I might pick Postgres.

Why a monolith? One developer, one database, shared data model. I'd split out the activity-ingest endpoint first when volume demands it.

Is the API key authentication? Yes — it's a bearer credential, not an identifier. Plain text, no expiry, no scope.

What if someone steals a key? They can fake your activity and mess up the leaderboard. They can't read your dashboard — that needs the login cookie.

How would you fix the key? Hash it at rest with an ID prefix, allow multiple per user, rotation with a grace period — or move to short-lived tokens / OAuth device flow.

Why not OAuth for the extension? A device flow is the right answer; the key was faster to ship. It's a legitimate MVP trade-off.

Why JWT and not sessions? Stateless — no session store to run, survives restarts. The cost is no easy revocation and no refresh token.

How does data get from the extension to the dashboard? Extension event → tracker counters → ~2-min flush → check API key → `$inc` **the numbers into the right 10-minute record** → (later) website asks → check JWT + ownership → read records → add up → JSON → chart.

Where's the aggregation done? Mostly in JavaScript after `Activity.find()`. Tech debt — it should be MongoDB `$group`. Streak, insights, the group leaderboard and the daily rollup already use `$group`.

What's the bottleneck? Still the leaderboard — it reads the whole activity table on every request. (Yesterday's 10-minute bucketing cut the row count ~10×, buying time, but the complexity class is unchanged.)

How do you scale the leaderboard to 10M users? A per-user `UserStats` rollup updated on write, then a Redis sorted set for O(log n) rank lookups.

Is Insights machine learning? No — plain statistics (averages, medians, a spread score), computed fresh each load. An LLM description layer is designed but not built.

Why statistics not ML? No labelled data, not enough users, and every number must be explainable.

Can users cheat the leaderboard? Easily — send fake hours with a valid key. There's no rate limit or validation. Fix: validate duration, rate-limit, require a unique send ID, cross-check with editor activity.

What happens if the same activity is sent twice? It `$inc`s the same 10-minute record again — so it double-counts *within* that bucket, but doesn't add a phantom row. Still not fully idempotent; proper fix is a client-generated ID per send + a unique index.

Two requests at the same time — any problem? Ingest is now an atomic `findOneAndUpdate` with `$inc` — two flushes for the same window both apply cleanly, and the unique bucket index handles the create race with a retry. The weak spots elsewhere are group-join (unique rule catches it but returns a 500) and adding a team member (a read-modify-write — should be `$addToSet`).

Why is `userId` a string on activities but an ID everywhere else? Historical. It forces type-conversion tricks in the leaderboard and blocks joins. Migrating it is on the list.

What breaks first at scale? The leaderboard (whole-table scan per request), then the serverless cron. The per-user analytics scans are now indexed and bounded by the bucketing.

Your frontend build — walk me through the cleanup. It had 29 old TypeScript errors (mostly unused imports, plus 5 implicit-any props in one decorative component). Fixed 2026-09-09; `npm run build` is green and CI keeps it that way. It was a cleanup, not a design issue.

How do you know the database queries are correct if nothing ran against a real DB? I don't fully — the pure functions are unit-tested (and the streak test runs the *shipped* code, not a copy), but there's no integration test. That's the biggest gap and it's on the plan.

What was the hardest part? Making the *time* number honest — idle time, and lines that net to zero on a refactor. I built an idle/pause state machine and switched to gross edit counts plus a churn metric.

What would you improve first? Hash the keys; add the error/build-success columns to the group leaderboard (the original motive — read-path change only); repoint the all-time reads at the daily-summary rollup that already exists; fix the TypeScript errors; add integration tests.

You changed the write model recently — what and why? It was one row per flush (~1 per minute of coding). I changed ingest to `$inc` into a 10-minute `(user, project, language)` record. `$inc` adds the real measured seconds, so every total is identical — only the time-of-day detail is now 10-minute-grained, which is the finest any chart shows. Row count dropped ~10×. There's a rollback env switch (`ACTIVITY_BUCKET_MS=0`).

15. "Defend your project" — the tough follow-ups

"Why MongoDB?" → "But you have a join table and embedded arrays — isn't that SQL fighting a document store?"

> *Partly true. The join table is me reaching for a relational pattern because groups need it, and it works — the unique index gives the guarantee. It's a sign the social half would be happier in Postgres. The activity half, which is the bulk and the hot path, genuinely suits documents.*

"Your leaderboard is O(n) per request." → "A cache fixes that."

> *A cache hides the cost behind a timer — the full scan still happens. The real fix is the rollup table, which changes the complexity. A cache on top of that is a nice extra.*

"Why not Redis / a queue / WebSockets?"

> *Nothing needs them yet. Leaderboard and insights finish in under a second at demo scale. Redis is the obvious next dependency (leaderboard + cache + rate limits). A queue earns its place past ~10k concurrent users. WebSockets I'd add for real-time group activity, not for a 30-second notification poll.*

"You could have shipped faster with Firebase."

> *True for the CRUD and auth. But the analytics are custom aggregations over a high-volume collection, and Firebase's per-document-read pricing would hurt — the daily chart alone reads a week of records. And the extension needs a stable server API regardless.*

"How do you stop cheating if a determined user spreads fake data thinly?"

> *Cross-check signals. A real coding hour has editor edits and focus events. Weight the leaderboard on corroborated activity, and cap how much any single send can contribute.*

"Nothing is tested against a real database — how do you trust it?"

> *I trust the pure logic (unit-tested, and the streak test runs the real shipped code). I don't fully trust the queries end-to-end — that's stated in my notes.* `supertest` *plus an in-memory MongoDB closes it and it's the top testing item.*

"If this got 10,000 real users tomorrow, what's your first move?"

> *Add the* `timestamp` *index and put a 30-day window on the leaderboard query as a stopgap, then build the* `UserStats` *rollup. In parallel: turn on* `helmet` *and rate limiting so ingest can't be flooded, and add validation so* `duration` *can't be faked. Then a health check and real logging.*

16. Final reminders

The 10 weaknesses to own before they're pointed out

1. **API key** — plain text, never expires, no limits.
2. **Leaderboard** — reads the whole table every request; no cache, no pagination; leaks emails.
3. **Analytics maths runs in JavaScript**, not the database — downloads records to add up (bounded now that they're bucketed, still not ideal).
4. ~~Cron breaks on serverless~~ ✅ fixed 2026-09-09 — `require.main` guard + an external trigger (`POST /api/internal/run-*` behind a secret) drives both the hourly sweep and the nightly rollup.
5. **Rate limiting is only on** `/auth` + `/api/extension` (group `/join` and analytics are still uncapped; no per-key ingest quota). `helmet` and ingest bounds-checking landed 2026-09-09.
6. ~~Frontend build fails~~ ✅ fixed 2026-09-09 (was 29 TypeScript errors).
7. **Fake data on the dashboard** ("Repeated Failures" panel); **unsaved to-dos**; **dead Teams page**.
8. **The group leaderboard doesn't show the error comparison yet** — the original motive; the data's collected, the view isn't built.
9. **No integration tests; nothing tested against a real database** — including the new bucketing/rollup code (pure-function tests only).
10. **All-time reads still scan raw** `activities` — not repointed at the `dailysummaries` rollup; the 400-day auto-delete is just a safety net.

(Fixed 2026-09-08: per-flush document explosion, the missing `timestamp` index, the dead `date` field, idle-<2-min inflating totals.)

For each one: know *why it matters*, *what it would take to fix*, and *why it's acceptable for a student project right now*.

Don't trip on these

- Don't call Insights "AI" or "ML". It's statistics. Say so first.
- Don't say the leaderboard is "optimised". It's a full scan. Know the rollup fix.
- Don't say the API key is "just an identifier". It's a credential.
- The frontend builds cleanly as of 2026-09-09 (CI enforces). It used to fail `tsc`.
- Don't claim integration test coverage. There is none against a real database.
- Don't claim the extension has an offline queue. It's memory-only.
- The dashboard "Repeated Failures" panel now shows real data (fixed 2026-09-09).
- There are 3 contributors — describe what *you* can explain in depth, don't over-claim.

Five sentences that make you sound senior

1. "The API key is basically a plain-text password with no expiry — a fine MVP, but I'd hash it at rest with an ID prefix and add rotation with a grace window."

2. "The leaderboard reads the whole activity table per request; the fix is a per-user stats rollup updated on write, turning a full scan into a 50-row read, with a Redis sorted set for rank lookups."
3. "I recently moved ingest from one row per flush to an atomic `$inc` into a 10-minute `(user, project, language)` bucket — real seconds are summed either way, so every total is identical, but the write and row rate dropped about 10×."
4. "The project's really about friendly competition — the group leaderboard is the point, and the next feature is surfacing the per-person error and build-success comparison, which the extension already collects."
5. "I audited my own code into an improvement plan — about 30 findings by severity — and closed the serious security ones with a test that fails if a protected route loses its auth check."